

Replication Certification at Publication— Checkable by Anyone, Anywhere, in Seconds, Forever*

Gary King[†] Yiqing Xu[‡] Leo Y. Yang[§]

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Abstract

Authors increasingly deposit replication datasets—the data and code behind published results—in archives such as Dataverse, and journals increasingly require it. But to the ever-skeptical scientific community a deposit alone proves little, so authors keep paying: months while a data editor reruns analyses, years answering readers’ questions, and rancor as failed replications become accusations because no record establishes what actually ran. We introduce a *certificate*: a cryptographically signed record of the deposited files and output captured by specialized logging software. It replaces the data editor’s rerun with evidence of the author’s own run that anyone can check, which no one, including the archive, can backdate, and no one can fake without risking detection. Certification asks little of anyone and benefits everyone: The author deposits, then reruns the analysis one extra time under an archive-issued random seed; the archive automatically hashes, signs, timestamps it in Bitcoin’s blockchain, and stores the record—confidential data included, under restricted access—but never runs the analysis; the journal simply checks a website that the certificate exists; and anyone can verify the record in seconds, or replay any step of it, even decades later.

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[†]Institute for Quantitative Social Science, Harvard University; [GaryKing.org](https://garyking.org), King@Harvard.edu.

[‡]Department of Political Science, Stanford University; yiqingxu.org, yiqingxu@stanford.edu.

[§]Stanford Center on China’s Economy and Institutions, Stanford University, leoyang@stanford.edu; currently affiliated with the Department of Accountancy, Economics and Finance, Hong Kong Baptist University; LeoYang.org, leoyang@hkbu.edu.hk.

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1. Introduction

Ensuring that evidence offered in a scholarly publication stands independently of its author is a goal that echoes through the history of science, back even to the 1660 motto of the Royal Society, *Nullius in verba* (“take nobody’s word for it”). We can also see this goal in the now widely accepted “*replication standard* [which] holds that sufficient information exists with which to understand, evaluate, and build upon a prior work if a third party could replicate the results without any additional information from the author,” (King, 1995, p.444), and in many efforts to meet the standard.¹

Scholarly publication outlets over four decades have adopted increasingly strict policies designed to convince authors to deposit *replication datasets*, the data and code behind a published result, in professional archives and to try to ensure each deposit provides meaningful evidence that results can be reproduced. Unfortunately, each of the policy changes has merely moved the burden of verifying replicability from one actor to another, while often increasing costs in money, staffing, publication delays, time investment by authors and future researchers, and misunderstandings between them, while still never quite succeeding. Section 2 explains.

Instead of even stricter and more expensive policies, we propose a technological solution to accomplish the same goals more effectively and at lower cost. By marshalling well-tested cryptographic methods, we show how to create a special replication dataset *certificate* that provides the evidence that the analysis ran on exactly its files and produced exactly the results it records, with its random draws made after the files were fixed. The journal can then substitute its expensive and time-consuming in-house replication runs with a quick check that the certificate exists before final acceptance, and a later reader can check the record in seconds instead of re-running the analysis. A failed replication then has a certified record to be argued from rather than a reputation to be argued about. See Section 3.

The rest of the paper states what certification does not cover (Section 4) and reviews what our proposal borrows from prior work and what that work leaves open (Section 5). The Supplementary Appendix provides all technical details, including what evading certification costs an attacker (Section S5.2) and our contributions (Section S6).

2. Problems with Replication Policies

No policy tried thus far lets anyone verify a deposited replication dataset without re-running it, so each future researcher has to verify the same deposit from scratch. We describe here the levels of enforcement journals have escalated through; the evidence indicating how these policies have failed; and the costs imposed by the policies in time, money, and rancor.²

¹We define “replication” as most users and archives do: running the same code on the same data, producing the same result. Some call this “reproduction” and reserve “replication” for a study redone with new data (National Academies of Sciences, Engineering, and Medicine, 2019); we use the words interchangeably.

²The norm that authors deposit the code and data behind a published result is long-established in the social sciences (Dewald, Thursby and Anderson, 1986; King, 1995), and professional archives now supply the storage, versioning, and persistent identifiers this protocol builds on (King, 2007). The Transparency and Openness Promotion guidelines extend the same levels across the sciences, with verified re-execution as their top level (Nosek et al., 2015).

2.1. Policy Enforcement Levels

Level 1 is the original and least stringent proposal, where journals require authors to “indicate in their first footnote in which public archive they will deposit the information necessary to replicate their numerical results.” Editors would be “responsible only for the existence of the footnote, not for confirming that the data set has been submitted nor for checking whether the results actually can be replicated.” Any verification “can and should be left to future researchers” (King, 1995).

Level 2 requires authors to go a step further and make data available on request, as in *Science*’s rule from 2011. But asked for the materials behind 204 papers published under it, only 44% of authors supplied them, and the results of only 26% of the 204 could be reproduced (Stodden, Seiler and Ma, 2018).

Level 3 makes replication dataset deposit a condition of publication, as now required by many leading outlets (Brodeur et al., 2026; Ferguson et al., 2023; Miske et al., 2026). In economics, the requirement reached the *American Economic Review* in 2005 and the *Quarterly Journal of Economics* in 2016 (Christensen and Miguel, 2018). In political science, the editors of twenty-seven journals signed the joint statement of the Data Access and Research Transparency initiative, committing to require that cited data be available in a trusted repository (Lupia and Elman, 2014; Data Access and Research Transparency (DART), 2015). Among quantitative political-science articles, a working link to the data and code was rare before about 2008, with the share growing approximately exponentially since (Rainey et al., 2025, fig.1).

Finally, Level 4 is re-execution of the code by the journal, originally as an option and then a requirement. From July 2009, *Biostatistics* let authors ask it to run their code and marked the article *R* if it reproduced; in two years, five of 125 articles earned the mark (Peng, 2011). Journals now increasingly require human-verified re-execution by the journal’s data editor (Vilhuber, 2020; Nüst and Eglen, 2021). The *American Journal of Political Science* has worked with the Odum Institute at the University of North Carolina since 2015 to verify every accepted quantitative article, postponing publication by seventy-five days on average (Jacoby, 2018).

2.2. Policy Failures

The literature suggests three conclusions. First, *archival deposit alone does not show that the code runs*. Of more than nine thousand R scripts deposited in Dataverse, 74% did not run to completion without error (Trisovic et al., 2022). Of 15,817 Python notebooks from biomedical publications with declared dependencies, 1,203 of those ran through without error (Samuel and Mietchen, 2024), in line with earlier mass executions of notebooks (Pimentel et al., 2019). When two teams each tried to redo one analysis from each of 18 microarray papers, ten could not be reproduced (Ioannidis et al., 2009). For all available replication packages for articles published 2000-2026, in major economics journals, only 11.2% 4,452 could be fully reproduced ().

Second, *running the code does not guarantee that the materials reproduce the article’s results*. Working only from deposited files, one audit reproduced the key result of just 22 of 67 published economics papers (Chang and Li, 2022). At *Cognition*, mandated deposit raised

the availability of usable data sharply, while 13 of 35 audited articles still had at least one value that could not be reproduced even with the authors’ help (Hardwicke et al., 2018). In the largest journal-run audit, 68% of attempts succeeded in replicating results (Fišar et al., 2024).

And finally, *journal verification catches more failures without catching them all*. Of the twenty-four papers given in-house replication review at the *Quarterly Journal of Political Science*, 14 of 18 produced results that differed from those in the manuscript (Eubank, 2016). Across 254 papers from the three leading political-science journals whose materials could be obtained, 61 did not reproduce every coefficient exactly and 20 did not reproduce it even approximately (Xu and Yang, 2026).

2.3. Policy Costs

Finally, journal replication policies impose three categories of costs.

The first cost is *time*, which comes in two parts, delayed acceptance and endless post-publication questions. In the former, median verification times at the American Economic Association’s journals have run three months per round, and some manuscripts need several rounds (Vilhuber, 2024); the average manuscript the *AER* published in 2025 spent thirteen weeks in the data editor’s hands, up from nine in 2022 (Luttmer, 2026). In the latter, authors assemble replication datasets to a data editor’s standard but still have to answer questions about them for years. Having access to a deposited replication dataset is far better than no deposit at all, but still considerable work: in one mass-reproduction project, teams averaged thirteen days per paper, and about one article in twelve required more than a hundred days (Brodeur et al., 2024). None of that work is recorded where the next reader can use it and so every researcher starts from scratch.

The second cost is *money*, paid by journals (and by editors in their time) which hire and manage replicators, in addition to providing cutting edge computational resources used by researchers. The American Economic Association’s Data Editor pays for all student employment and pre-doctoral staff, while Cornell University’s Economics Department provides tuition waivers and computing (Vilhuber, 2025). The Cornell laboratory that replicates received, in one year, 529 requests covering 415 manuscripts. About 15-20 undergraduate research assistants work there per week, supervised by the three staff (Vilhuber et al., 2022). At the *American Journal of Political Science*, verification by the Odum Institute took an average of eight person-hours per manuscript, and authors resubmitted materials 1.7 times on average, a “not trivial” financial cost (Jacoby, Lafferty-Hess and Christian, 2017). *Management Science* charges \$79 on every original submission since 2025, to fund its reproducibility infrastructure and external verification (INFORMS, 2025).

The final cost is *rancor*. When a replication fails, the absence of a shared record of what was actually run often turns technical questions into personal ones. The dispute over Reinhart and Rogoff’s result turned on what had actually been run—which spreadsheet, which exclusions, which weighting—and was settled only when the authors found and shared the original file (Herndon, Ash and Pollin, 2014). The dispute over LaCour and Green’s article turned on whether the deposited data had been collected at all (Broockman, Kalla and Aronow, 2015). Daniel Kahneman, acting as a senior statesman, has gone so far as to propose etiquette rules for how readers attempting replications should approach authors

(Kahneman, 2014).

3. A Technological Solution: The Certificate

Figure 1 gives an overview of the certification protocol. The first three parts of this section follow the figure’s first three panels: how certification works (Section 3.1); why it is sufficient evidence (Section 3.2); and what everyone gains (Section 3.3). We save discussions of confidential data until Section 3.4; the figure’s fourth panel summarizes what certification does not cover (Section 4).

3.1. How It Works

Certification has three parts corresponding to the three bands within Panel 1 of Figure 1. First, the author makes the replication dataset **READY** by locking down all sources of variability (due to the code, computing environment, or settings), so that it gives the same results if run repeatedly. Second, the author and archive work together to **CERTIFY** the replication dataset. And finally, anyone, at any point in the future, can **CHECK** the results in seconds without having to rerun the analysis, and each check catches frauds that pass the one before it.

3.1.1. Ready: Author Prepares the Replication Dataset

The author uses our certificate software to ensure their replication dataset gives the same results when run twice on their machine, hence defining “replicability” for this study. The software first collects (a) every file in the replication dataset, with its hash, and the results of a *dependency audit*, a reading of the scripts for the files they open and the seeds they set; (b) the settings to freeze, such as the time zone and the number of processor cores (Section S2.2); and (c) the software environment itself (deposited as a “container image”, a packaged copy of the operating system and programs, where software licenses allow, and otherwise as add-on libraries installed inside the replication dataset; Section S2.1). From these, the software creates a *closed copy*, which includes the replication dataset plus the small files added by the certificate software. The closed copy enables all the author’s analyses to be run with one top-level command, keeps track of each step in the process, and, where possible, has only one input for a random seed.

Finally, the author conducts two runs of the closed copy, what we call the *closure runs*, each from a fresh temporary copy and under a seed of the author’s choosing. The certificate software compares every file the two runs wrote and grades their agreement on a scale of six *determinism tiers*, from byte-for-byte identical, through equal within a stated numeric tolerance, to a real difference or a run that did not finish (Table S1). The dataset is *ready* when the two runs agree to within half a unit in the fourth significant figure of every number, though the goal is byte-identical agreement; the software writes a report, deposited with the replication dataset, that names any file that differed (Section S2.2).



Figure 1. Certification at a 1. Panel 1: **READY**, the author prepares the replication files until two runs agree. **CERTIFY**, the archive issues a seed for the fingerprinted files, the author runs the analysis under it, and the archive signs and timestamps the record; draws fall after the seed is issued and before the certificate is issued. **CHECK**, three options, depending on need. Panel 2: why the record is evidence. Panel 3: what each party pays **TODAY** and gains once **CERTIFIED**. Panel 4: what certification does not cover.

3.1.2. Certify: Author Deposits, Receives Seed, Runs Once; Archive Seals

Certification is designed to eliminate the possibility of *backdating* by anyone including even the archive (in order to surreptitiously change the record after the fact), *seed shopping* (where the author cherry-picks a supposedly random seed to make their results look better), *substitution* (replacing an intermediate file, such as the cleaned data, after the run while claiming the certificate still covers it), and *ambiguity* (in what results the run produced or how to conduct a new one). To take a dataset that is “ready” and certify it to receive these benefits requires three steps (corresponding to the Certify row of Panel 1 in Figure 1, with details in Section S3).

First, the author *deposits* the closed copy in the archive. The archive then computes a new random seed with three inputs: a hash of the uploaded files, the archive’s secret cryptographic key, and a recent Bitcoin block — meaning that no one, the author included, could know the seed before it is computed. The archive’s receipt, its signed and timestamped statement that these files arrived, records the hash, the block, and the seed (Section S3.1). Readers cannot recompute the seed without the archive’s secret key, so they check the archive’s signature on the receipt instead (Section 3.2).

Second, the author *runs* the closed copy once under the seed issued by the archive (producing the “certified run”). For each step in the run, the certificate software records the “pre-state” (every file and its hash present before the step starts), any files it writes, and any output it produces. Because each step begins exactly where the previous one ended, every pre-state already includes the hashes of all files written by earlier steps, so the record of each step is tied to everything that came before it. The software then combines the records of all steps, in order, into a single hash so that changing any intermediate file, or dropping, adding, or reordering a step, would change this hash. The certificate therefore vouches not only for the final numbers but for the entire path from the closed copy to the published results. The published numbers must come from this run (Section S3.2). The software produces a “proposed certificate”.

Finally, the archive *seals* the proposed certificate submitted by the author. To do this, the archive checks the certificate, refuses on any failure, signs the “witness” (its statement that it sealed this certificate, using its signing key), timestamps the certificate on the blockchain, and enters the witness in its ledger (a public list of every document it issues including this one, kept so that any later edit is visible publicly). The archive stores the certificate and records it in a metadata field linked to the published dataset (Section S3.3; Section S8 lists everything an archive must do).

3.1.3. Check: Anyone Can Easily Verify the Record

Anyone can check a certificate on their own computer, without taking anyone else’s word for what was run or when (Section S4). Three types of checks are available, depending on need; after describing them, we show that each catches frauds that pass the one before it.

Three Types of Checks. First, the *existence check* enables a journal to confirm that the replication dataset, the article’s citation of it, and the certificate exist, and that the certificate carries no exception mark (the archive’s flag indicating that the certificate falls

short of one of the protocol’s standards). This requires only a fast lookup on the archive’s webpage; no reanalysis is necessary (Section S4.3). An existence check is sufficient for the journal because a sealed certificate enables anyone, without help from the journal, to verify both “record consistency” (that the stored files are those the run started from and wrote, and that neither the files nor the record has changed since) and “replication” (that rerunning the recorded steps from their stored starting files produces outputs that agree with the certified ones within the certificate’s determinism tier).

Second, the *record check* is designed for someone who prefers not to take the archive’s webpage on trust. Here, our open-source certificate software recomputes every hash from the files and checks the archive’s signatures, the timestamps, and the ledger, validating record consistency. The record check runs no analysis, typically takes from seconds to a few minutes, needs no outside service when run offline, and, as long as the cryptography it uses stays infeasible to break, works the same way decades from now (Sections S4.1 and S5.1.4).

Finally, the *re-execution check* is designed for a future researcher who wishes to build on this research or to test whether the record was faked. The researcher can rerun the entire analysis and see whether its results agree with the certified ones within the determinism tier. A rerun takes as long as the original, is one command where a container image was deposited, and is judged only when the researcher’s software environment matches the recorded one (Section S4.2); a rerun that agrees gives the researcher a tested starting point for new work. Conveniently, a researcher can also rerun a single step, without running any of those that came before or after, whenever all of that step’s starting files are stored; the certificate reports how many steps qualify.

Each Check Catches Frauds That Evade the One Before. A certificate is evidence of only record consistency and replication. By all accounts failure to replicate is far more often due to honest error than fraud (Section S5.2), but we aim for certification to hold up even against an author acting in bad faith (in security terms, a “threat actor”). We also consider cases where an author and archive collude to produce a fraudulent result (which is presumably even rarer because we know of no such case). We now summarize how each type of certification check catches frauds that pass the one before it, and the permanent public record reaches what passes all three; Section 4 states what certification does not cover.

First is *passing the existence check, caught by the record check*. The existence check relies on the archive’s webpage. If it is faked, the record check catches the fraud and any file changed after the certified run, such as edited output tables, swapped data files, or secretly rerun scripts (Section S4.1).

Second is *passing the record check, caught by re-execution*. To commit fraud while passing the record check, an author must falsify the record itself: alter the analysis while it runs, such as through hardware control of the computer or a value changed in memory; edit the record before the archive seals it; or hack the certificate software so it still reports the genuine copy’s hash (Section S3.1). A colluding archive could also sign a falsified record and add it to its ledger. In each case the stored starting files remain untouched, so a reader who reruns a falsified step with a genuine copy of the software, in an environment matching the recorded one, will obtain the honest result and discover the fraud (Section S4.2). Each additional researcher who reruns the analysis (which is likely to happen more often with

more important scholarly claims) increases the probability of detection.

Last is *passing the re-execution check, caught by the evidence trail*. Permanent public access protects science from its worst actors, whose frauds reproduce on every rerun or lie outside the run, and so pass all three checks. The certified replication dataset, its certificate, and the archive’s ledger stay publicly available, and no one can backdate them. Anyone may read code that fabricates results, or that reads a file or sets a seed the dependency audit missed (Section S5.1), and examine the data, which a certificate does not claim are genuine, as Broockman, Kalla and Aronow (2015) did. Anyone may also count the deposits made to draw a new seed, each a ledger entry, and compare printed numbers with certified ones, as we rely on the copyeditor or author to do (Section S4.3).

3.2. Why it Needs No One’s Word

Unlike articles published under prior replication policies, a certified paper carries its own evidence, which anyone can check, anywhere, in seconds, forever, without taking anyone’s word for what was run or when: not the author’s, not the journal’s, not even the archive’s. *Evidence* here means a record anyone can check, which no one, including the archive, can backdate, and no one can fake without risking detection. This is summarized in four features (corresponding to Panel 2 of Figure 1).

First, *replication dataset files are fixed before the certification run*. The archive hashes the deposit and computes the seed from that hash, so the seeded draws cannot precede the files; this rests on well-tested cryptography, not on trust (Section S1). Seed shopping therefore needs a new deposit for each seed, and the archive’s ledger publicly records every deposit (Section S5.1).

Second, *the results are fixed after the certification run*. The certificate records the hash of every file each step the code started from and produced and every logged estimate, and the witness names the certificate’s own hash. Thus, a later change to any file or to the record fails the check.

Third, *the certification run is fixed in time*, meaning that a researcher trying to correct earlier mistakes will fail or leave a visible trail. The receipt and the certificate are each timestamped in Bitcoin’s blockchain, so the certified run’s random draws are bounded between two block times. The later edge is the time of the block that timestamps the certificate. The earlier edge is the time of the block the receipt names, since the seed is computed from that block’s hash (Section S3.3.2). Not even the archive can backdate either timestamp, and that earliest date is what no signature, log, or timestamp could supply on its own (Section 5).

Finally, *the result is checkable by anyone*. No special permission is required. Everything else the checks test is arithmetic, which anyone can easily recompute starting from the replication dataset and the certificate. The archive’s word is taken only for two items: that it issued the seed as specified (Section S7) and the facts only it holds (who deposited and whether a revised deposit and an earlier deposit it names as its predecessor have the same uploader; see Section S3.4). That the run happened as recorded rests on the open-source certificate software, which any reader can test by re-running recorded steps (Section S4.2).

3.3. How Everyone Benefits

Certification lowers the cost of verification by changing who runs the analysis and how often: the author runs it once more, on their machine that already holds their software environment, and everyone afterward checks the record instead of rebuilding and rerunning it from scratch. Panel 3 of Figure 1 shows the gains party by party. Because several gains reach more than one party, we take them here by the problem in Section 2 that each one solves, with what certification costs in the same terms. Those problems are the failures of Section 2.2 and the three costs of Section 2.3, taken in that order.

First, *policy failures*. Current replication datasets, even when required by journal policies, carry no evidence that their code runs or reproduces the article’s results, and journal verification catches only some issues (Section 2.2). In contrast, a certified replication dataset carries evidence that its analysis ran on exactly its files and produced exactly the results it records, which the copyeditor or author checks during copy editing to connect the certificate to the publication (Section S4.3). Because checking time does not grow with the length of the computation (Section S4.1), the evidence extends to analyses too long for a data editor to re-execute, which happen because published articles often represent the computational cutting edge. The closure runs, of the kind authors already make before submission anyway, are now deposited and fixed in the record, where readers can test them, while today no one else ever sees an author’s own runs.

Second, *time*. Data editors presently must rebuild the author’s software environment, rerun their analyses, and ask the author for whatever is missing. This takes months per round before acceptance, each later researcher rebuilds it all from scratch, and authors answer questions about their deposits for years (Section 2.3). Certification replaces publication delays and later questions with one more run by the author at final acceptance, on the machine that already holds the hardware, software, libraries, and data, so that the run comes down to nearly one command. A future reader or researcher attempting a replication confirms that the record is intact with the record check, which is fast and requires no copy of the author’s software environment (Section 3.1.3), and answers from the record the questions about what was run that would otherwise go to the author. Confirming that a step still produces its recorded outputs takes that step’s own running time (where the container image was deposited; without an image, the reader must also install the recorded software versions; Section S4.2). The author pays in making one extra run; a revision that changes the files after certification repeats all three runs (Section S3.4).

Third, *money*. Journals today pay for the data editors, staff, and computing that re-execution requires (Section 2.3). For a certified paper, journal staff only need an existence check (webpage lookup) at final acceptance with no technical choices (or special expertise) required, because the protocol fixes the standard every certificate must meet. Archives pay to operate the software that performs the deposit and seal steps, and for storage of the record, which includes the files the run writes and so can exceed the deposit itself (Section S3.6). The archive’s institutional work of custody, versions, backups, metadata, terms of use, restricted access, and indefinite storage continues, as before.

Finally, *rancor*. Today a failed replication becomes a dispute about what the author did, because no shared record says what was run (Section 2.3). Certification gives the scientific community a public record of precisely what was run for every certified paper, so a dispute

over a failed replication turns on what the record shows, which both sides can check, instead of on what the author says was run. The publication and its certified replication dataset then stand independently of the person who created them. Disputes will continue, but only about future research, as should be the case.

3.4. Extensions to Confidential Data

We now extend our certification protocol to confidential data, those requiring access levels that are not fully public, such as requiring click-through licenses, guestbook signatures, signed contracts, or committee approvals. We distinguish three types by who is permitted to hold the files: the archive, only a secure facility, or only the author under a license. We assume the archive may publish the data’s metadata regardless (otherwise no one would know the data existed in the first place). The guarantees weaken in that order, as the files move further from the archive’s custody.

First, *restricted data the archive is allowed to hold* require only trivial changes to the protocol (Section S3.5). The certificate includes no data values, and our summaries of the data are *salted* hashes, which are hashes of the data mixed with a secret random value so that no one can confirm a guess about a file’s contents. Before release, the author verifies that the certificate does not include the identifies of a person or organization in the files, so that it may be made public. A later reader must then satisfy the archive’s access rules for this dataset before checking the files.

Second, *data only accessible inside a secure facility*, such as a federal research data center, is not reachable by the archive, but its notary work needs only the hashes, so the facility takes over the work that uses the files. The author makes the closure runs and the certified run inside the facility, and only salted hashes, the seed, and a certificate whose estimates pass the facility’s review for disclosure leave it. The archive computes the seed from the salted hashes; facility staff run the record check on the files and sign the result, which the archive verifies before sealing; and the facility keeps the exact files and software environment for a stated period. The certificate is made public. Anyone can then check the signatures, dates, ledger, and deposited code, and a reader admitted to the facility following the same rules as the author did can check the data or rerun the analysis. The cost here is trust in the facility’s signed check, and checks that last only as long as the facility keeps the files.

Finally, *data licensed under terms that forbid archival deposit*, such as from commercial vendors, are analyzed on the author’s computer as usual and enter the deposit only as salted hashes and metadata naming the source and version (such as restricted data used in recent papers in the American Economic Association’s journals; Vilhuber 2024). Everything but the data is certified in full, and the seed still follows the data’s hashes, but only the author checks the data before sealing. A later reader can check them only by obtaining the same license and version of the data, which vendor is trusted to maintain; the archive then supplies the salt file needed to compare the hashes.

4. What Certification Does Not Cover

A certificate is evidence of two claims, record consistency and replication (Section 3.1.3). What lies outside them forms four groups, as follows (Section S5).

First, *a certificate is evidence about a computation, not about the science*: not that the analysis answers the paper’s question, avoids *p*-hacking, is free of model dependence, holds in fresh data, or contains genuine data. It locks down one run from the deposited files forward, at final acceptance; changed files need a new certificate, linked to the earlier one (Section S3.4). It dates the record and the random draws, not the substantive work (Section S3.3.2); binds output files and logged estimates, not the printed numbers which is left to the author and journal copyeditor (Section S4.3); and shows which account deposited the files in the archive, not who wrote them.

Second, *practical limits bound which analyses can be certified*. One is the size of the files the archive will accept: a data file larger than the archive will store enters only by its hash and download address, and rerunning a step that reads it requires finding a copy with that hash (Sections S2.3 and S4.2). Another is the number of files the archive will hash separately: a folder of millions of separate small files, such as one per social-media post, would need millions of hashes, so it is hashed as a single unit instead. Data too large or restricted to deposit, and a remote service’s answers, such as a large language model or other API, can be saved to an intermediate file, with only the analysis from that file forward certified; and data the archive may not hold carry weaker guarantees (Section 3.4).

Third, *how evidence ages*. The record check works as long as the archive’s files survive and the archive re-timestamps them in time if Bitcoin is ever abandoned; re-execution works only while the recorded software can be run, after which the software would need to be rewritten (Section S5.1.4). A deposited container image holds the installed software, not a list of libraries that may later disappear, but it runs only on the kind of processor it was built for, such as Intel’s or Apple’s, or in emulation software. Software whose license forbids deposit, such as Stata, cannot go in the image.

Finally, *what a bad actor can still do*. Frauds that pass all three checks (from Section 3.1.3) remains on the permanent public record at risk of being found (Section 3.1.3). This includes results fabricated in the code; code that reads a file or sets a seed the dependency audit misses; and seed shopping paid for with a new, recorded deposit per seed. Re-execution catches a falsified step only when a researcher reruns it (Section S4.2). In addition, the archive is trusted to issue seeds as specified: a colluding archive, or an author who steals the archive’s secret key, can make future seeds predictable without a record; and a stolen signing key alone yields certificates the ledger lacks, which the record check reveals against a later copy of the ledger the archive cannot revise.

5. Prior Work

We build on three bodies of prior work: (1) designs that give evidence about a published analysis; (2) the tools the protocol borrows; and (3) methods that date work by drawing public randomness after a commitment. (Section S6 summarizes our contributions.)

First, the designs, in order of increasing assurance. File fingerprints show whether a file changed (The Dataverse Project, 2026), and verifiable computational results store each published result under an identifier the article cites (Gavish and Donoho, 2011). Signed provenance, graded by the framework SLSA (Supply-chain Levels for Software Artifacts), signed execution records that authors can neither alter nor disown, and platforms such as Whole Tale that package the environment for re-execution rest on the word of whoever signed

or packaged (Open Source Security Foundation, 2026; Keita and Homan, 2026; Brinckman et al., 2019). TRACE, whose goal is similar to ours, requires its own platform to run and attest to the workflow, but this lets it reach data usable only inside secure facilities (Vilhuber et al., 2023; Li et al., 2025). Journal data editors, the named checkers of CODECHECK, and the French certification agency cascadi re-run the analysis themselves, at the cost of a qualified person per paper (Nüst and Eglen, 2021; Pérignon et al., 2019). Trusted hardware signs output produced inside a protected processor, on the word of the chip’s maker (Costan and Devadas, 2016; Li, 2026); cryptographic proofs of computation need no such trust but remain far too costly today for general data analysis (Goldwasser, Micali and Rackoff, 1989; Walfish and Blumberg, 2015). A certificate needs neither a platform that vouches for the run nor a second party’s re-run.

Second, the borrowed tools. The cryptography is well-tested: Merkle trees, keyed hashes, digital signatures, and public timestamps (Section S7), obtained in Bitcoin through OpenTimestamps (Todd, 2016), with a ledger modeled on the append-only logs that Certificate Transparency keeps for website certificates and Sigstore for signed software, an internet standard (Laurie, Langley and Kasper, 2013; Newman, Meyers and Torres-Arias, 2022; Birkholz et al., 2026). Blockchain timestamps of research files and pipelines exist, but show only that files existed by some time (Gipp, Meuschke and Gernandt, 2015; Wittek et al., 2021; Casas and Fidler, 2024). The recording of a run draws on running twice under frozen settings and comparing the outputs (Lamb and Zacchiroli, 2022; Bizopoulos and Bizopoulos, 2020), and on signed per-step records of inputs and outputs and the provenance workflow engines keep (Torres-Arias et al., 2019). Re-executing a random sample of recorded steps comes from distributed computing (Monrose, Wyckoff and Rubin, 1999; Du et al., 2004; Haeberlen et al., 2010; Jia et al., 2021); the deterrence random checks provide has been analyzed as well (Golle and Mironov, 2001; Barnett, Zardo and Graves, 2018).

Finally, dating work by public randomness drawn after a “commitment”, a public statement that fixes the materials. Election audits (Cordero, Wagner and Dill, 2006; Lindeman and Stark, 2012), the Internet Engineering Task Force’s selection of committee members (Eastlake, 2004), randomness beacons (Rabin, 1983; Kelsey et al., 2019), and a randomization seeded from a Bitcoin block (Saa and Stern, 2019) all draw their randomness after publishing such a commitment, and a signature can postdate a blockchain’s latest block, which dates the signature rather than a run’s draws (Abadi et al., 2020). Seeds hashed from the commitment itself (Fiat and Shamir, 1987) or from a preregistration (Mazor, Mazor and Mukamel, 2019), and blind analysis, which hides results under a perturbation of the data until the analysis is frozen (MacCoun and Perlmutter, 2015), leave trust to whoever commits it, a gap that running the whole study once inside protected hardware closes, at the price of ruling out re-execution (Meißner et al., 2021). A certificate needs no separately published commitment.

6. Concluding Remarks

A certificate is designed to move the scientific community closer to the idea of keeping evidence in publications based solely on the record and independent of the author. It turns one additional run of the analysis by the author already prepared to do so, made under the archive’s seed after deposit, into evidence that anyone can check, and it lets a journal

replace its own rerun with a simple look-up on a webpage. We suggest the journal print the replication dataset’s usual citation, by DOI and version, with one word added, “Certified” (Section S4.3). The supplementary appendix gives the more specific specification for version 1 of our protocol, from how an author makes a dataset ready (Section S2) to the byte-level rules an implementer needs (Section S9).

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